

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Analyse and Practise

Name: _____

Date: _____

LEARNING INTENTION	investigate how scientific knowledge informs responses and decisions by individuals and communities
KEY IDEA	Science provides evidence about likely effects, but decisions also consider cost, access, values, uncertainty and local knowledge. Good reasoning separates evidence from preference.
WORKED MODEL	Temperature maps show where shade is lowest. A school compares trees, sails and roof coatings, then weighs cooling evidence, water use, cost and accessibility.

SUCCESS CRITERIA - I CAN

[] identify evidence [] compare responses [] recognise trade-offs [] justify a decision

A. NOTICE AND IDENTIFY

1. Sort as evidence or consideration: measured temperature, installation cost, personal favourite colour, maintenance data.

2. Explain why one measurement is insufficient for a community decision.

3. Identify the scientific question in a decision about reducing playground heat.

B. APPLY AND EXPLAIN

4. Compare three responses using effectiveness, risk, cost and equity.

5. Write one claim the evidence supports and one it cannot support.

6. Explain how uncertainty should be communicated rather than hidden.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Build a decision matrix with four criteria and evidence notes.

8. Review it for groups who may be affected differently.

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D. APPLY THE SKILL IN CONTEXT

9. Data: trees reduce surface temperature 8 C, sails 11 C, pale coating 5 C. List questions needed before choosing.

10. Recommend one option or combination and justify with evidence plus two considerations.

11. State what new evidence could change the recommendation.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate: 'The option with the largest temperature reduction is automatically best.'

13. Draft a 90-word community recommendation separating finding, trade-off and decision.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[] I can use scientific evidence and trade-offs to justify a community decision.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult decision was _____ because _____

One correction or improvement I made: _____

My next practice step: _____

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Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. evidence; consideration/economic evidence; preference; evidence.
2. Conditions vary; repeated measurements and comparable methods are needed.
3. Example: Which response safely reduces heat most effectively in this setting?
- 4-8. Require criteria, evidence and affected groups, not preference alone.
9. Cost, lifespan, coverage, water, safety, biodiversity, access and measurement conditions.
10. Any justified option using data and two relevant considerations.
11. Examples: long-term trials, cost, shade area or maintenance evidence.
12. False; effectiveness is one criterion among risks, feasibility and equity.
13. Must distinguish evidence, uncertainty/trade-off and recommendation.

TEACHER CHECK AND NEXT STEP

[] Uses accurate science ideas and evidence.

[] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

Science informs decisions but does not remove values or practical constraints.

Do not present uncertain evidence as proof; state limits and conditions.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.